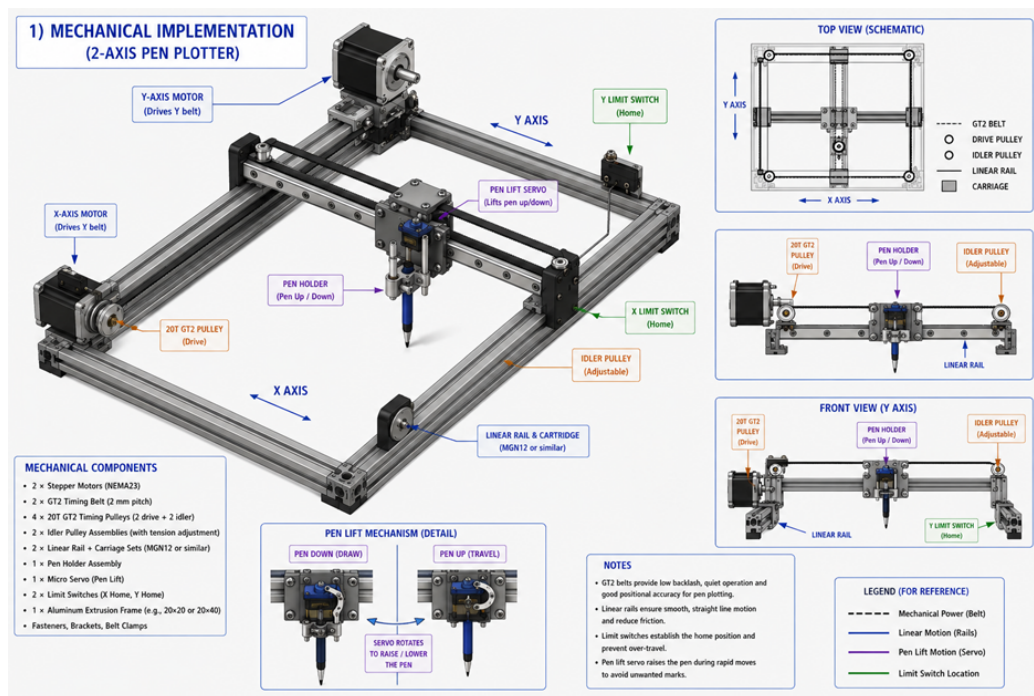


Pen Actuation & Motion Control

Task 4.1 – CNC Pen UP/DOWN Mechanism Without a Z-Axis Motor



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0. Executive Summary

The plotter frame used for this training carries two stepper axes (X and Y) and a Ruida RDC6445S(EC) controller, but no Z-axis motor. This report designs the mechanism that lifts and lowers the pen without one, using an output the controller already has: the laser-enable line on its CN5 interface. Section 1 restates the problem and the electrical starting point. Section 2 explains the chosen approach end to end, from the controller's output pin to the physical motion of the pen. Section 3 proposes and verifies the real components, with prices and sources. Section 4 gives the full build: wiring, power, and the truth table linking the controller's output state to Pen UP / Pen DOWN. Section 5 compares this design against two other ways of solving the same problem and explains why this one was chosen.

1. Understanding the Problem

What is available. The machine has X and Y stepper motion only; there is no third motor to raise or lower the tool. The controller is a Ruida RDC6445S(EC), a board built for laser cutting/engraving rather than milling, which means it already exposes a dedicated laser-control interface (connector CN5) even though this build has no laser. That interface is the “available CNC controller output” the task description points to.

What CN5 actually provides. The controller's laser interface carries two relevant signals: L-ON1, a TTL enable line, and LPWM1, an analog power-control line. Electrically, the controller's laser output is a modulated signal running at 2–50 kHz with a 0–5 V swing and a 1–99% duty-cycle range, the TTL side can source/sink up to 50 mA, and the inputs/outputs are already galvanically isolated inside the controller. L-ON1 specifically is the enable line: for a glass-tube-style laser output driven in low-level form, this pin is the one that switches the laser's power supply on and off – exactly the simple binary flag a pen-lift mechanism needs, with no laser hardware required to make use of it.

Why a servo cannot be wired to CN5 directly. L-ON1 is a clean two-state (ON/OFF) signal, not a continuous waveform. A hobby servo, however, needs to continuously receive a 50 Hz pulse train (roughly a 1–2 ms pulse repeated every 20 ms) in order to hold any angle at all – if the pulse train stops, the servo goes limp. A single on/off flag cannot hold a servo in position by itself, so something has to sit between L-ON1 and the servo to keep generating that 50 Hz pulse train and to decide, moment to moment, which of the two pen positions it should represent.

The design task, restated. Turn the controller's L-ON1 flag (drawing / not drawing) into a continuously-held servo position (pen down / pen up), using the smallest, cheapest, most reliable piece of hardware that can do that translation, and confirm every stage is electrically compatible with the RDC6445S.

2. Proposed Solution

Chosen approach: a small microcontroller reads L-ON1 and continuously drives a micro servo, which pushes a cam against a spring-loaded pen carriage. The controller's own G-code already produces the correct flag for free: RDWorks-generated laser-mode output turns L-ON1 active whenever a cutting/drawing move is in progress and inactive on travel moves – the same ON-while-drawing, OFF-while-traveling behavior a Z-axis pen-lift needs, just arriving on a different pin than a Z motor would use. An Arduino Nano watches that one pin and holds the correct servo angle on its own output pin at all times, regardless of what the RD controller is doing at that instant. The servo turns a small lever/cam a fixed number of degrees; the cam pushes a spring-loaded pen holder down against a return spring when the pen must draw, and releases it – letting the spring lift the pen – when the machine is only traveling.

Why this method was selected. It needs no firmware modification to the RD controller (some pen-plotter builds instead reflash GRBL to output true servo pulses on a spindle pin – not applicable to a Ruida board, and not desirable on a controller also used for other jobs). It reuses a channel the controller already isolates internally, so no extra opto-isolator is needed purely for safety. And it keeps the mechanical side identical to the one already planned in the system architecture (Section D1 of the pre-training research): a low-voltage pen-lift servo powered from its own regulated supply, exactly as already listed in that bill of materials – this design fills in the one detail that was left open, namely what sits electrically between the controller and that servo.

Complete signal path.

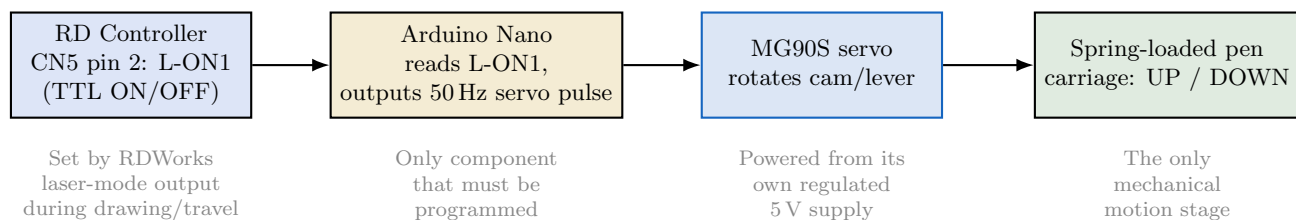


Figure 1: End-to-end signal path from the controller's laser-enable output to the physical Pen UP / Pen DOWN motion. The Arduino Nano is the single translation stage that converts a two-state flag into a continuously-held servo position.

3. Proposed and Verified Components

Component	Function & key specs	Price (Mikro-Electron, Amman)	Source / Link
Arduino Nano V3.0 (CH340, ATmega328P)	Reads the isolated L-ON1 flag and continuously generates the 50 Hz / 1–2 ms servo pulse for the current pen state. 5 V logic, one digital input + one PWM-capable output pin used	JD 4.45 ($\approx$ \$6.30)	MikroElectron, SKU ME-2767 https://mikroelectron.com/product/me-2767
TowerPro MG90S metal-gear micro servo	Lifts/lowers the pen via the cam. 2.0 kg-cm torque @4.8 V, 2.8 kg-cm @6 V, 180° rotation, 4.8–6 V operating range, metal gear train	JD 4.90 ($\approx$ \$6.91)	MikroElectron, SKU ME-6399 https://mikroelectron.com/product/me-6399
LM2596 DC–DC step-down (buck) module	Steps the machine’s 24 V motor rail down to a clean 5 V supply dedicated to the servo and the Nano, matching the “servo power supply / DC-DC converter” line already in the system BOM. Adjustable 1.3–35 V out, up to 3 A	JD 2.35 ($\approx$ \$3.32)	MikroElectron, SKU ME-244 https://mikroelectron.com/product/me-244
10 k Ω metal-film resistor (1/4 W, 1%)	Holds the Arduino’s input pin at a known logic level between L-ON1 transitions (CN5’s L-ON1 is an open-collector style output)	JD 0.03 ($\approx$ \$0.04)	MikroElectron, SKU ME-265 https://mikroelectron.com/product/me-265
Male–male / male–female jumper wires (qty $\sim$ 10–12)	Interconnects Nano, servo, DC–DC converter, and CN5; sold individually or in ribbon strips	$\sim$ JD 1.00 for 10 pcs	MikroElectron, jumper-wire range https://mikroelectron.com/category/Connectors-Wires
Return spring (light compression/extension)	Pulls the pen carriage back to the UP position the instant the servo releases the cam	\$1–2	Not stocked by MikroElectron – hobby/hardware store
3D-printed pen-holder carriage + cam bracket	Slides vertically in a short guide rail on the tool head; the servo horn drives a cam or pushrod against the carriage	\$2–5 in filament, or printed on-site	Custom STL; MikroElectron also lists in-house 3D-printing as a service https://mikroelectron.com/services
M3 hardware (screws, nuts, standoffs)	Mounts the servo bracket and carriage guide rail to the tool head	\$1–2	Not stocked by MikroElectron

Table 1: Full bill of materials for the pen-lift mechanism

3.1 Electrical Compatibility Check

CN5 to Arduino Nano. The controller’s laser TTL output swings 0–5 V and can source or sink up to 50 mA, comfortably inside the Arduino Nano’s 5 V-tolerant digital input rating, which draws negligible

current as an input. The pull-up resistor guarantees a defined logic level even in the instant before or after L-ON1 switches. Because CN5's outputs are already galvanically isolated inside the RDC6445S, no separate opto-isolator module is required purely for protection – that isolation is already built into the controller for this specific interface.

Arduino Nano to MG90S servo. A standard hobby servo's signal wire expects a 5 V logic pulse, which the Nano's digital output pins supply directly – no driver chip or level shifter needed. The MG90S's rated operating voltage (4.8–6 V) matches the DC-DC converter's 5 V output exactly.

Power budget. The MG90S can draw up to roughly 500 mA under load, which the Nano's own onboard 5 V regulator is not sized to supply on top of running the microcontroller itself. For that reason the servo is powered from the dedicated DC-DC converter rather than the Nano's 5 V pin, with grounds tied together between the Nano, the servo, the DC-DC converter, and the RD controller's signal ground (pin 1 of CN5) so all logic levels are referenced to the same 0 V.

Note on the LM2596 module's stock status. At the time of writing, MikroElectron's LM2596 step-down module (ME-244) shows as out of stock. Any equivalent LM2596-based buck module (or the in-stock adjustable step-down modules under MikroElectron's "DC-DC Converters" category) is a drop-in substitute, since the design only requires a clean, adjustable 5 V output at low current.

4. How to Build and Connect It

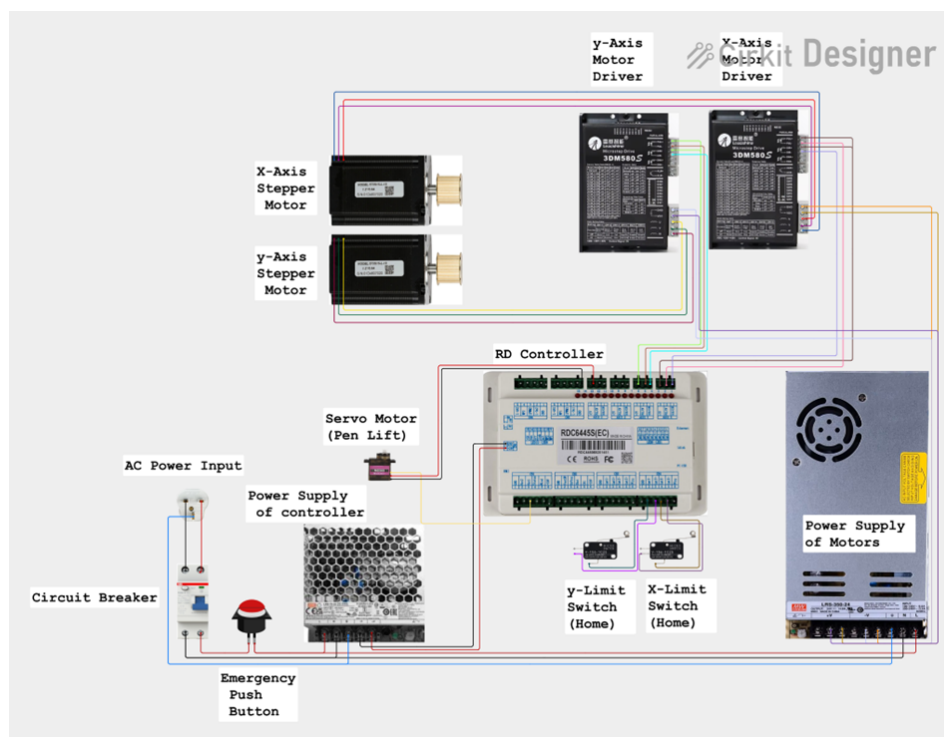


Figure 2: Full-machine wiring reference from the system architecture (Section D2 of the pre-training research)

4.1 Comprehensive Wiring Diagram

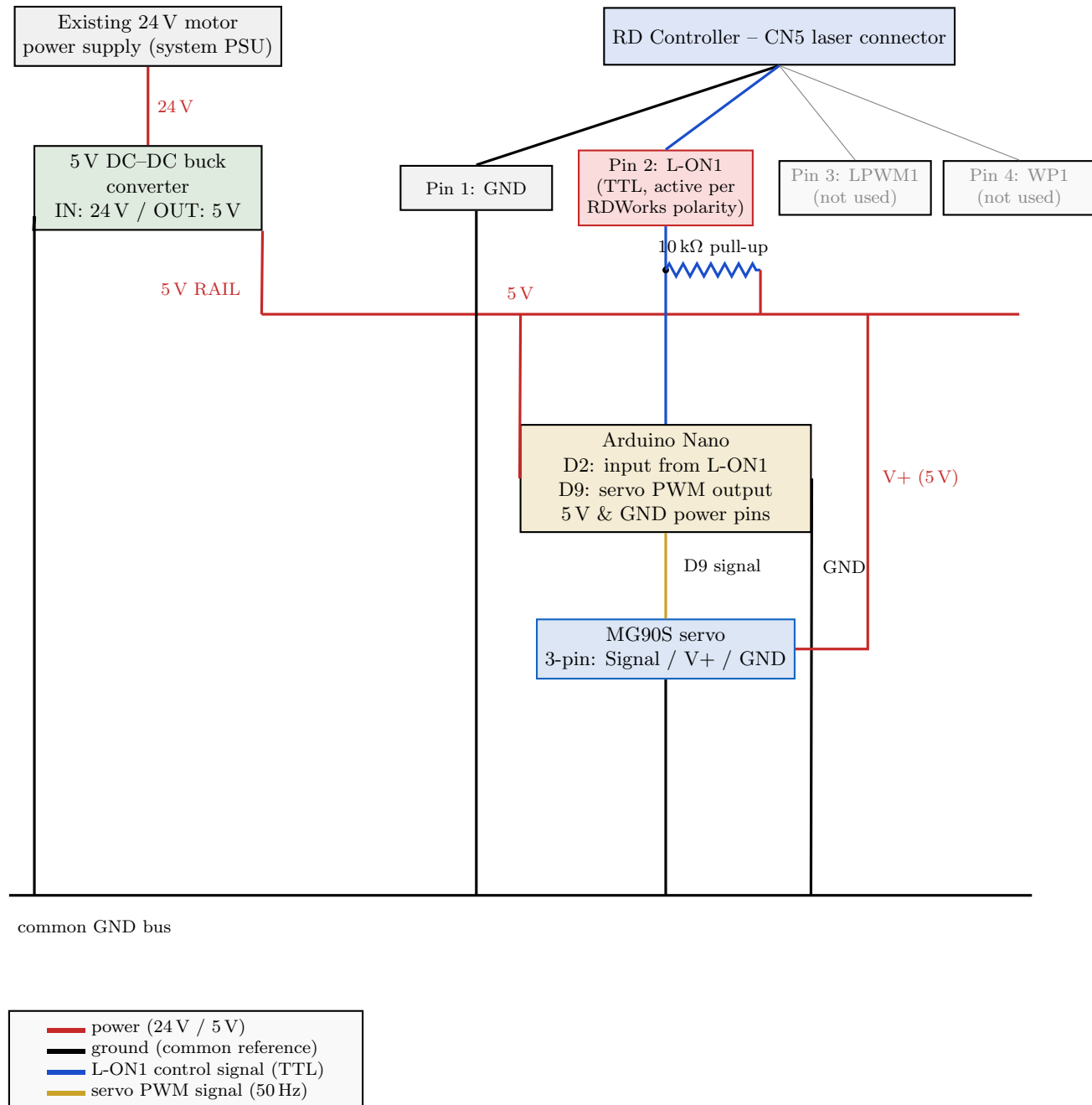


Figure 3: Comprehensive wiring diagram: from the existing 24 V system power supply, through the 5 V buck converter, the RD controller's CN5 connector (all four pins shown, with pins 3 and 4 unused), the pull-up resistor and Arduino Nano, to the MG90S servo. All grounds – buck converter, controller CN5 pin 1, the Nano, and the servo – run in their own dedicated columns down to a single common ground bus, as required whenever isolated and non-isolated stages share a signal reference.

4.2 Operation Truth Table

RDWorks state	CN5 L-ON1	Nano output	Physical result
Drawing move in progress	Active (laser-enable asserted)	Servo commanded to DOWN angle	Cam pushes the carriage down against the spring; pen tip touches the paper
Travel move (no drawing)	Inactive	Servo commanded to UP angle	Cam releases; the spring pulls the carriage back up, lifting the pen clear of the paper

Table 2: How the controller’s laser-enable state maps to Pen UP / Pen DOWN through the Nano and servo. Exact DOWN/UP angles are set once in the Nano’s sketch and tuned to the physical cam geometry and desired pen pressure.

4.3 Mechanical Sketch: Pen Carriage in Both States

5. Comparison of Alternatives

Approach	What it needs	Trade-off
MCU + servo (chosen)	Arduino Nano reading L-ON1, generating a 50 Hz servo pulse	Precise, repeatable pen-down force; angles adjustable in software; needs one extra programmable part, but that part is cheap and simple to build/debug
555 timer + servo	Analog astable circuit; L-ON1 switches between two RC networks (via a diode/transistor) to select pulse width, no code required	No programming needed, but the UP/DOWN angles are set by trimpots that drift with temperature and component tolerance – harder to guarantee consistent pen pressure over a long drawing
Solenoid + MOSFET	A single logic-level MOSFET switches solenoid power directly from L-ON1 (via a level/current buffer); a flyback diode protects the MOSFET	Simplest circuit by far, no MCU at all, but only a hard on/off “punch” – no control over pen pressure, higher current draw, and the mechanical snap risks nudging the light XY carriage on every stroke

Table 3: Alternatives considered against the chosen MCU + servo design.

Why the MCU + servo design was selected. Drawing quality depends on the pen touching down with the same, controlled force every single time – a solenoid’s abrupt mechanical snap is more likely to jitter the pen tip or disturb the light belt-driven XY carriage than a servo’s smooth, controlled sweep, and the 555 timer’s drift makes it difficult to guarantee that same pen-down force stroke after stroke.

The Arduino Nano adds one small, well-understood part in exchange for angles that are set once in software, easy to re-tune if the cam geometry changes, and stable over the length of a drawing job.

6. Summary & Checklist

The mechanism uses the RD controller's existing CN5 laser-enable output (L-ON1) as a ready-made drawing/travel flag, requiring no firmware changes to the controller itself. An Arduino Nano translates that two-state flag into a continuously-held 50 Hz servo pulse, since a servo cannot hold a position from an on/off signal alone. The MG90S servo drives a cam against a spring-loaded pen carriage, giving a controlled Pen DOWN when drawing and a spring-retained Pen UP when traveling, powered from a dedicated 5 V DC–DC converter already anticipated in the system's bill of materials. Checklist for the build:

- Confirm L-ON1's active polarity in RDWorks' laser-mode output settings before wiring the pull-up direction.
- Wire CN5 pin 1 (GND) to a common ground shared by the Nano, the DC–DC converter, and the servo.
- Power the servo from the DC–DC converter, not the Nano's onboard 5 V pin.
- Tune the Nano's UP/DOWN servo angles to the physical cam geometry once assembled, checking that the pen makes consistent contact without excessive downforce.
- Test with a repeated shape (as in the earlier stepper-accuracy test) to confirm the pen lifts cleanly on every travel move with no drag marks.

7. References

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